

Advanced Log Analysis Report

1. Introduction

Advanced log analysis is a fundamental function of a Security Operations Center (SOC) that enables analysts to detect complex cyber threats by examining security logs generated from multiple systems such as endpoints, firewalls, applications, and network devices. Modern cyberattacks often involve multiple stages, including unauthorized access attempts, privilege escalation, and data exfiltration, which may not be detectable through isolated logs.

This task focuses on implementing log correlation, anomaly detection, and log enrichment techniques using **Elastic Security** and Security Onion tools. By correlating Windows authentication logs with network traffic events, suspicious activities such as brute-force attacks and abnormal data transfers can be identified. The activity demonstrates how SOC analysts use centralized logging and behavioral analysis to detect threats, reduce false positives, and improve incident detection accuracy.

The methodology aligns with industry best practices recommended by **SANS Institute**, **MITRE ATT&CK threat framework**, and incident investigation case studies published by **Cybersecurity and Infrastructure Security Agency**.

2. Objective

The objective of this task is to analyze and correlate system and network logs to identify suspicious activities that may indicate cyberattacks.

Key Objectives

- Correlate logs from multiple sources to detect attack patterns.
 - Identify failed login attempts using Windows Event ID 4625.
 - Detect anomalous network behavior such as high-volume outbound traffic.
 - Enrich logs with contextual information such as IP geolocation.
 - Reduce false positives through contextual analysis.
-

3. Tools Used

- Elastic Security - Log ingestion, correlation, and anomaly detection
 - Security Onion - Network monitoring and security analysis
 - Google Sheets - Log documentation and analysis
 - GeoIP Plugin - Log enrichment with location data
-



4. Methodology / Implementation

4.1 Log Correlation

Log correlation involves combining logs from different sources to identify relationships between events that indicate malicious activity.

Sample Windows authentication logs containing Event ID 4625 were ingested into Elastic Security. These logs represent failed login attempts that may indicate brute-force attacks or credential stuffing.

Network logs were then analyzed to identify outbound connections occurring after repeated login failures. The correlation process revealed patterns suggesting potential system compromise.

Log Correlation Table

Timestamp	Event ID	Source IP	Destination IP	Notes
2026-02-18 12:00:00	4625	192.168.110.140	8.8.8.8	Suspicious DNS request
2026-02-18 12:02:00	4625	192.168.110.140	8.8.4.4	Multiple failed logins
2026-02-18 12:05:00	4624	192.168.110.140	Internal Server	Successful login after failures
2026-02-18 12:07:00	Network Event	192.168.110.140	External Host	Unusual outbound traffic

Correlation Findings

- Repeated authentication failures were detected.
- A successful login occurred after multiple failed attempts.
- Suspicious outbound traffic was initiated from the same host.
- This behavior indicates possible credential compromise and command-and-control communication.

4.2 Anomaly Detection

An anomaly detection rule was created in Elastic Security to detect abnormal network activity.

Detection Rule

- Condition: `bytes_out > 1MB` within 1 minute
- Purpose: Identify potential data exfiltration or malware communication.



A simulated file transfer exceeding normal baseline traffic triggered the rule successfully, generating an alert in the SIEM dashboard.

Anomaly Detection Observations

- Sudden spikes in outbound traffic were detected.
- The behavior deviated from normal user activity patterns.
- Alerts helped identify potential data exfiltration attempts.

4.3 Log Enrichment

Log enrichment enhances raw log data by adding contextual information.

A GeoIP plugin was used to enrich IP address data with geographic location and risk information.

Enrichment Example

IP Address	Country	Risk Indicator
192.168.110.140	Internal Network	Low Risk
203.45.67.21	Unknown External Region	Suspicious

Enrichment Summary

GeoIP enrichment provided additional context to network logs by identifying the geographic origin of IP addresses. External connections from unusual locations were flagged as suspicious. Enrichment improved threat visibility, enabled better risk assessment, and supported faster decision-making during security investigations.

5. Results and Findings

The advanced log analysis process identified several suspicious behaviors:

- Multiple failed authentication attempts from a single host.
- Successful login after repeated failures indicating possible credential compromise.
- High-volume outbound traffic suggesting data exfiltration.
- External connections to unknown IP addresses.
- Improved detection accuracy using log enrichment.
-

Log correlation significantly reduced false positives by providing contextual relationships between events.



6. Security Impact

Advanced log analysis improves SOC capabilities by:

- Detecting multi-stage cyberattacks.
 - Improving incident response efficiency.
 - Identifying compromised accounts.
 - Enhancing threat visibility.
 - Strengthening organizational security posture.
-

7. Conclusion

This task demonstrated the implementation of advanced log analysis techniques in a SOC environment using Elastic Security and Security Onion. Through log correlation, anomaly detection, and log enrichment, suspicious system behavior was successfully identified and analyzed. The exercise highlights the importance of centralized logging, contextual analysis, and behavioral monitoring in detecting complex cyber threats and improving incident detection accuracy in modern security operations.